
■ CASE 10.3 STEPS TO SUCCESS

Janet Richards fixes her eyes on those of her partner Gilbert Baker and says firmly, “All right. Let’s do it.”

And with those words, InterCat, a firm founded by Janet and Gilbert that specializes in the design and maintenance of Internet catalogs for small consumer businesses, will be going public. InterCat employs 30 individuals, with the majority of

them computer programmers. Many of the employees have followed the high-technology market very closely and have decided that since high-technology firms are more understood and valued in the United States than in other countries, InterCat should issue its stock only in the United States. Five million shares of InterCat stock will comprise this new issue.

The task the company has ahead of itself is certainly daunting. Janet and Gilbert know that many steps have to be completed in the process of making an initial public offering. They also know that they need to complete the process within 28 weeks because they need the new capital fairly soon to ensure that InterCat has the resources to capture valuable new business from its competitors and continue growing. They also value a speedy initial public offering because they believe that the window of opportunity for obtaining a

good stock price is presently wide open—the public is wild about shopping on the Internet, and few companies offering Web page design services have gone public.

Because the 28-week deadline is breathing down their necks, Janet and Gilbert decide to map the steps in the process of making an initial public offering. They list each major activity that needs to be completed, the activities that directly precede each activity, the time needed to complete each activity, and the cost of each activity. This list is shown below.

Activity	Preceding Activities	Time	Cost
Evaluate the prestige of each potential underwriter.		3 weeks	\$ 8,000
Select a syndicate of underwriters.	Evaluate the prestige of each potential underwriter.	1.5 weeks	\$ 4,500
Negotiate the commitment of each member of the syndicate.	Select a syndicate of underwriters.	2 weeks	\$ 9,000
Negotiate the spread* for each member of the syndicate.	Select a syndicate of underwriters.	3 weeks	\$12,000
Prepare the registration statement including the proposed financing and information about the firm's history, existing business, and plans for the future.	Negotiate both the commitment and spread for each member of the syndicate.	5 weeks	\$50,000
Submit the registration statement to the Securities and Exchange Commission (SEC).	Prepare the registration statement.	1 week	\$ 1,000
Make presentations to institutional investors and develop the interest of potential buyers.	Submit the registration statement to the SEC.	6 weeks	\$25,000
Distribute the preliminary prospectus affectionately termed the red herring.	Submit the registration statement to the SEC.	3 weeks	\$15,000
Calculate the issue price.	Submit the registration statement to the SEC.	5 weeks	\$12,000
Receive deficiency memorandum from the SEC.	Submit the registration statement to the SEC.	3 weeks	\$0
Amend the registration statement and resubmit it to the SEC.	Receive deficiency memorandum from the SEC.	1 week	\$ 6,000
Receive registration confirmation from the SEC.	Amend the registration statement and resubmit it to the SEC.	2 weeks	\$0

Activity	Preceding Activities	Time	Cost
Confirm that the new issue complies with the “blue sky” laws of each state.	Make presentations to institutional investors and develop the interest of potential buyers. Distribute the preliminary prospectus affectionately termed the red herring. Calculate the issue price. Receive registration confirmation from the SEC.	1 week	\$ 5,000
Appoint a registrar.	Receive registration confirmation from the SEC.	3 weeks	\$12,000
Appoint a transfer agent.	Receive registration confirmation from the SEC.	3.5 weeks	\$13,000
Issue final prospectus that includes the final offer price and any amendments to all purchasers offered securities through the mail.	Confirm that the new issue complies with the “blue sky” laws of each state. Appoint a registrar and transfer agent.	4.5 weeks	\$40,000
Phone interested buyers.	Confirm that the new issue complies with the “blue sky” laws of each state. Appoint a registrar and transfer agent.	4 weeks	\$ 9,000

*The spread is the payment an underwriter receives for services.

Janet and Gilbert present the list of steps to the employees of InterCat. The head of the finance department, Leslie Grey, is fresh out of business school. She remembers the various project management tools she has learned in business school and suggests that Janet and Gilbert use PERT/CPM analysis to understand where their priorities should lie.

- (a) Draw the project network for completing the initial public offering of InterCat stock. How long is the initial public offering process? What are the critical steps in the process?
- (b) Janet and Gilbert hear through the grapevine that their most fierce competitor, Soft Sales, is also planning to go public. They fear that if InterCat does not complete its initial public offering

before Soft Sales, the price investors are willing to pay for InterCat stock will drop, since investors will perceive Soft Sales to be a stronger, more organized company. Janet and Gilbert therefore decide that they want to complete the process of issuing new stock within 22 weeks. They think such a goal is possible if they throw more resources—workers and money—into some activities. They list the activities that can be shortened, the time the activity will take when it is fully shortened, and the cost of shortening the activity this much. They also conclude that partially shortening each activity listed below is possible and will give a time reduction and cost proportional to the amounts when fully shortening.

Activity	Time	Cost
Evaluate the prestige of each potential underwriter.	1.5 weeks	\$14,000
Select a syndicate of underwriters.	5 weeks	\$ 8,000
Prepare the registration statement including the proposed financing and information about the firm’s history, existing business, and plans for the future.	4 weeks	\$95,000
Make presentations to institutional investors and develop the interest of potential buyers.	4 weeks	\$60,000

Activity	Time	Cost
Distribute the preliminary prospectus affectionately termed the red herring.	2 weeks	\$22,000
Calculate the issue price.	3.5 weeks	\$31,000
Amend the registration statement and resubmit it to the SEC.	5 week	\$ 9,000
Confirm that the new issue complies with the "blue sky" laws of each state.	5 week	\$ 8,300
Appoint a registrar.	1.5 weeks	\$19,000
Appoint a transfer agent.	1.5 weeks	\$21,000
Issue final prospectus that includes the final offer price and any amendments to all purchasers offered securities through the mail.	2 weeks	\$99,000
Phone interested buyers.	1.5 weeks	\$20,000

How can InterCat meet the new deadline set by Janet and Gilbert at minimum cost?

- (c) Janet and Gilbert learn that the investment bankers are two- timing scoundrels! They are also serving as lead underwriters for the Soft Sales new issue! To keep the deal with InterCat, the bankers agree to let Janet and Gilbert in on a little secret. Soft Sales has been forced to delay its public issue because the company's records are disorganized and incomplete. Given this new information, Janet and Gilbert decide that they can be more

lenient on the initial public offering timeframe. They want to complete the process of issuing new stock within 24 weeks instead of 22 weeks. Assume that the cost and time to complete the appointment of the registrar and transfer agent are the same as in part (b). How can InterCat meet this new deadline set by Janet and Gilbert at minimum cost?

Note: A data file for this case is provided on the CD-ROM for your convenience.

■ CASE 9.3 PROJECT PICKINGS

Tazer, a pharmaceutical manufacturing company, entered the pharmaceutical market 12 years ago with the introduction of six new drugs. Five of the six drugs were simply permutations of existing drugs and therefore did not sell very heavily. The sixth drug, however, addressed hypertension and was a huge success. Since Tazer had a patent on the hypertension drug, it experienced no competition, and profits from the hypertension drug alone kept Tazer in business.

During the past 12 years, Tazer continued a moderate amount of research and development, but it never stumbled

upon a drug as successful as the hypertension drug. One reason is that the company never had the motivation to invest heavily in innovative research and development. The company was riding the profit wave generated by its

hypertension drug and did not feel the need to commit significant resources to finding new drug breakthroughs.

Now Tazer is beginning to fear the pressure of competition. The patent for the hypertension drug expires in 5 years,¹ and Tazer knows that once the patent expires, generic drug manufacturing companies will swarm into the market like

¹In general, patents protect inventions for 17 years. In 1995, GATT legislation extending the protection given by new pharmaceutical patents to 20 years became effective. The patent for Tazer's hypertension drug was issued prior to the GATT legislation, however. Thus, the patent only protects the drug for 17 years.

vultures. Historical trends show that generic drugs decreased sales of branded drugs by 75 percent.

Tazer is therefore looking to invest significant amounts of money in research and development this year to begin the search for a new breakthrough drug that will offer the company the same success as the hypertension drug. Tazer believes that if the company begins extensive research and development now, the probability of finding a successful drug shortly after the expiration of the hypertension patent will be high.

As head of research and development at Tazer, you are responsible for choosing potential projects and assigning project directors to lead each of the projects. After researching the needs of the market, analyzing the shortcomings of current drugs, and interviewing numerous scientists concerning the promising areas of medical research, you have decided that your department will pursue five separate projects, which are listed below:

Project Up	Develop an antidepressant that does not cause serious mood swings.
Project Stable	Develop a drug that addresses manic-depression.

Project Choice	Develop a less intrusive birth control method for women.
Project Hope	Develop a vaccine to prevent HIV infection.
Project Release	Develop a more effective drug to lower blood pressure.

For each of the five projects, you are only able to specify the medical ailment the research should address, since you do not know what compounds will exist and be effective without research.

You also have five senior scientists to lead the five projects. You know that scientists are very temperamental people and will work well only if they are challenged and motivated by the project. To ensure that the senior scientists are assigned to projects they find motivating, you have established a bidding system for the projects. You have given each of the five scientists 1000 bid points. They assign bids to each project, giving a higher number of bid points to projects they most prefer to lead. The following table provides the bids from the five individual senior scientists for the five individual projects:

Project	Dr. Kvaal	Dr. Zuner	Dr. Tsai	Dr. Mickey	Dr. Rollins
Project Up	100	0	100	267	100
Project Stable	400	200	100	153	33
Project Choice	200	800	100	99	33
Project Hope	200	0	100	451	34
Project Release	100	0	600	30	800

You decide to evaluate a variety of scenarios you think are likely.

- Given the bids, you need to assign one senior scientist to each of the five projects to maximize the preferences of the scientists. What are the assignments?
- Dr. Rollins is being courted by Harvard Medical School to accept a teaching position. You are fighting desperately to keep her at Tazer, but the prestige of Harvard may lure her away. If this were to happen, the company would give up the project with the least enthusiasm. Which project would not be done?
- You do not want to sacrifice any project, since researching only four projects decreases the probability of finding a breakthrough new drug. You decide that either Dr. Zuner or Dr. Mickey could lead two projects. Under these new conditions with just four senior scientists, which scientists will lead which projects to maximize preferences?
- After Dr. Zuner was informed that she and Dr. Mickey are being considered for two projects, she decided to change her bids. The following table shows Dr. Zuner's new bids for each of the projects:

Project Up	20
Project Stable	450
Project Choice	451
Project Hope	39
Project Release	40

Under these new conditions with just four scientists, which scientists will lead which projects to maximize preferences?

- Do you support the assignment found in part (d)? Why or why not?
- Now you again consider all five scientists. You decide, however, that several scientists cannot lead certain projects. In particular, Dr. Mickey does not have experience with research on the immune system, so he cannot lead Project Hope. His family also has a history of manic-depression, and you feel that he would be too personally involved in Project Stable to serve as an effective project leader. Dr. Mickey therefore cannot lead Project Stable. Dr. Kvaal also does not have experience with research on the immune systems and cannot lead Project Hope. In addition, Dr. Kvaal cannot lead Project Release because he does not have experience with research on the

cardiovascular system. Finally, Dr. Rollins cannot lead Project Up because her family has a history of depression and you feel she would be too personally involved in the project to serve as an effective leader. Because Dr. Mickey and Dr. Kvaal cannot

lead two of the five projects, they each have only 600 bid points. Dr. Rollins has only 800 bid points because she cannot lead one of the five projects. The following table provides the new bids of Dr. Mickey, Dr. Kvaal, and Dr. Rollins:

Project	Dr. Mickey	Dr. Kvaal	Dr. Rollins.
Project Up	300	86	Can't lead
Project Stable	Can't lead	343	50
Project Choice	125	171	50
Project Hope	Can't lead	Can't lead	100
Project Release	175	Can't lead	600

Which scientists should lead which projects to maximize preferences?
(g) You decide that Project Hope and Project Release are too complex to be led by only one scientist. Therefore, each of these projects will be assigned two scientists as project leaders. You

decide to hire two more scientists in order to staff all projects: Dr. Arriaga and Dr. Santos. Because of religious reasons, the two doctors both do not want to lead Project Choice. The following table lists all projects, scientists, and their bids.

	Kvaal	Zuner	Tsai	Mickey	Rollins	Arriaga	Santos
Up	86	0	100	300	Can't lead	250	111
Stable	343	200	100	Can't lead	50	250	1
Choice	171	800	100	125	50	Can't lead	Can't lead
Hope	Can't lead	0	100	Can't lead	100	250	333
Release	Can't lead	0	600	175	600	250	555

Which scientists should lead which projects to maximize preferences?

- (h) Do you think it is wise to base your decision in part (g) only on an optimal solution for an assignment problem?

Note: A data file for this case is provided on the CD-ROM for your convenience.